CAO TIEN DUNG

EMBEDDED SYSTEMS & AI ENGINEER

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PROFILE

Senior Computer Systems Engineering student at Da Nang University of Science and Technology (DUT) with a proven dual foundation in Embedded Systems/IoT and Artificial Intelligence/Computer Vision. Developer of multiple production-ready AI applications, secure biometric locks, and microcontroller integrations. Seeking a professional internship to leverage software engineering, real-time hardware debugging, and edge AI deployment skills to build intelligent, high-performance systems.

TECHNICAL SKILLS

PROGRAMMING

C/C++ (OOP) · Python · MATLAB · Data Structures · Pointers & Memory

EMBEDDED & HARDWARE

STM32 (CubeIDE) · ESP32 · Arduino · Altium Designer · Proteus · PCB Design · Keil C · UART/SPI/I2C

AI & COMPUTER VISION

PyTorch · OpenCV · YOLOv8 · EfficientNet · Grad-CAM++ · EasyOCR

WEB & INFRASTRUCTURE

FastAPI · Streamlit · Docker · DVC · SQLite · Git

EDUCATION

2022 -> (Mar-Sep)/2027

Da Nang University of Science and Technology (DUT)

Bachelor of Engineering in Computer Systems

Faculty of Electronics & Telecommunications

Cumulative GPA: 3.15 / 4.0

LANGUAGES

English

TOEIC 650 (Intermediate, technical reading)

Japanese

JLPT N4 (Pursuing N3, conversational)

CERTIFICATIONS

KEY ENGINEERING PROJECTS

Tourism & Shopee E-Commerce Simulator — 2022 - 2023

gh https://github.com/caotiendung111/PBl1-Quanlitourdulich.2022-2023

C++ · OOP Concepts · Data Structures · Pointers · File Systems

Developed a C++ terminal application simulating real-life travel reservation flows and Shopee catalog transactions.

- Designed robust class architectures utilizing composition, inheritance, and dynamic pointer allocation to master fundamental memory boundaries in standard C++.

OCL Audio Power Amplifier Circuit Design — 2024 - 2025

gh https://github.com/caotiendung111

Altium Designer · Proteus · PCB Routing · Soldering · Analog Circuits

Researched, designed, and manufactured a physical Output Capacitor-Less (OCL) high-fidelity analog audio power amplifier.

- **Hardware Design:** Simulated analog stages inside Proteus, completed PCB schematics, and routed double-sided traces using Altium Designer.
- **Manufacturing:** Etched, drilled, and hand-soldered discrete components onto the physical board, executing rigorous thermal management procedures.
- **Validation:** Verified signal stability with oscilloscopes, obtaining high signal-to-noise ratio output with zero cross-over distortion.

AI-Based Automatic Sorting Conveyor System | PBL4 — 2025

gh https://github.com/caotiendung111/duanbangchuyen_nhom1_hethongnhung_AI_FIREBASE

C/C++ (OOP) · STM32 · ESP32 · OpenCV · Motor Control · Firebase

Engineered an industrial-scale automated sorting assembly belt connecting microcontrollers and computer vision modules.

- **Microcontroller Core:** Designed the master scheduler on an STM32 MCU written in C++ object-oriented style, ensuring precise servo/DC motor synchronization for sorting actions.
- **Edge Vision Integration:** Developed a communication channel between the STM32 controller and a local computer processing OpenCV camera feeds to scan QR/barcodes and categorize assets.
- **Cloud Ecosystem:** Connected ESP32 modules to sync sorting tallies to a Firebase Realtime Database and handled automated banking QR payment triggers.

FireGuard IoT — AI Smart Fire Alarm System — 2025 - 2026

gh https://github.com/caotiendung111/Du-An-Bao-Chay-IOT-AI-WEBSEVER1

ESP32 · YOLOv8 · Python · Streamlit · Telegram API · IoT Sensors

Invented a smart fire inspection framework bridging local embedded sensor nodes and visual deep learning modules.

- **Sensor Logic:** Configured ESP32 microcontrollers to interface with flame and smoke sensors, managing automated physical alarm actuators (buzzers, water pumps).
- **False Alarm Control:** Programmed a local temporal 10-frame noise filtering routine on visual feeds, dropping false smoke reports by over **90%**.
- **Remote Communication:** Deployed automated Telegram alerts with live capture images and GPS location data upon fire validation.

- **C/C++ Programming** - Codelearn
- **Python Programming** - Coursera

TrafficAI — LPR & Smart Vehicle Monitoring System
2025 - 2026

gh https://github.com/caotiendung111/AI-Traffic-Intelligence-Revenue-Ecosystem_MonTriTueNhanTao

YOLOv8 | OpenVINO | EasyOCR | FastAPI | SQLite | PWA

Programmed a commercial-grade smart parking solution tracking vehicle counts and recognizing license plates instantly from real-time network streams.

- **Neural Pipeline:** Fine-tuned a custom YOLOv8 detector combined with EasyOCR, optimized through Intel OpenVINO runtime, ensuring ultra-low latency inference on edge CPU architectures.
- **Dashboard Architecture:** Built a complete control dashboard with FastAPI and PWA support, handling complex search, soft-delete data bins, blacklist configurations, and visual audit logs.
- **Defense Shield:** Wrote custom middleware for IP-based rate limiting, WebSocket validation, and secure authentication to defend against malicious API flooding.

DeepLock Pro — Advanced Biometric Face Authentication
2026

gh https://github.com/caotiendung111/BioGuard_Face_Security_Mon_Tri_Tue_Nhan_Tao

OpenCV | face_recognition | AES-128 Fernet | CLAHE AI | Streamlit

Developed a premium, real-time facial verification suite equipped with robust presentation attack detection (liveness analysis) and biometric database encryption.

- **Liveness Analysis (Anti-Spoofing):** Implemented an active defense mechanism computing continuous Eye Aspect Ratio (EAR) across 15 frames alongside real-time Head Pose Estimation (Yaw/Pitch/Roll) to completely restrict print and screen replay attacks.
- **Biometric Security:** Extracted 128-D face descriptors and deployed encrypted database storage utilizing symmetric 128-bit AES Fernet cryptography with automated migration scripts.
- **Image Enhancement:** Integrated an adaptive light analyzer running Contrast Limited Adaptive Histogram Equalization (CLAHE) in the LAB color space, doubling face-matching accuracy in dark environments.

DeepGuard — AI-Powered Deepfake Detection System
2026

gh https://github.com/caotiendung111/DeepGuard-Deepfake-Detection

PyTorch | EfficientNet-B4 | FastAPI | Grad-CAM++ | Docker | DVC

Engineered a state-of-the-art production-ready deep learning pipeline designed to inspect and identify manipulated digital facial images and video records.

- **Machine Learning:** Trained an EfficientNet-B4 classification framework, achieving a benchmark **0.985 AUC-ROC** and **0.962 F1-score** on joint FaceForensics++ and Celeb-DF test sets.
- **Explainable AI (XAI):** Integrated Captum LayerGradCam algorithms to overlay heatmaps, pinpointing specific tampered facial areas to increase output interpretability.
- **DevOps & Serving:** Managed data version control utilizing DVC, developed high-performance REST endpoints with FastAPI, and containerized the pipeline via Docker Compose for uniform scaling.